

Rutgers University
Master's Comprehensive Examination

PART II

Applied (90 minutes)

December 7, 1991

Instructions: Attempt any four out of the following five questions. Mention which four you would like to have graded. Open book and open notes.

1) A researcher seeks to compare AZT (a new drug for treating AIDS) with a placebo. The researcher takes 200 individuals with AIDS who live in New York, and randomly divides them into two groups of 100 individuals each. The first group are treated with AZT, the second group with the placebo. A coresearcher working in Paris replicates the experement with 200 individuals with AIDS who live in Paris. After a month, each patient is evaluated as having improved or not having improved. The data is as follows:

	NEW YORK		PARIS	
	AZT	Placebo	AZT	Placebo
Improved	50	10	50	40
Not improved	50	90	50	60

statistically analyze the above data and summarized your conclusions.

2) A health insurer reviews bills that have been submitted by health providers. Of a total of 1000 bills submitted, the insurer draws a sample of 100 bills (by means of a simple random sample without replacement) and for each bill determines whether the amount billed was too much or too little, and the amount of the overcharge or undercharge. For the sample of 100 bills, the insurer finds that 20 of the bills were too high, 10 were too low, and 70 were exactly right. For the 20 bills that were too high, the mean overcharge was \$30, with a standard deviation of \$10. For the 10 bills that were too low, the mean undercharge was \$6 with a standard deviation of \$4.

- For all 100 bills, what is the mean of the overcharge.
- For all 100 bills, what is the standard deviation of the overcharge
- Give a 95% confidence interval for the proportion of bills that are too high.
- If you wanted to estimate the proportion of bills that are too high, within 0.01 of the true value with 95% confidence, how big a sample of bills should you take.

3) A researcher seeks to determine whether varying certain factors of the production process affects the hardness of a ceramic material. Once a batch of material has been composed, one-ounce balls of the material are taken from the batch and formed into ceramic "cookies." The cookies are pressed in a mold, and the first factor to be varied is the amount of pressure used in pressing the ceramic cookies. Four different levels of pressure are to be compared: 60 psi, 70 psi, 80 psi, or 90 psi. The ceramic cookies are then placed in an oven and baked for a certain amount of time and at a certain temperature. The second factor is the amount of time of baking which is varied at two levels: 30 minutes or 40 minutes. The third factor is the temperature of the oven which is varied at three levels: 300 degrees (F), or 350 degrees(F) or 400 degrees(F). After the ceramic cookies are baked, they are taken out of the oven, allowed to cool, and finally their hardness (the response variate y) is measured. There are $4 \times 2 \times 3 = 24$ treatment combinations.

The experiment proceeds as follows: A batch of material is made up and 24 ceramic cookies are made (and numbered from 1 to 24). The 24 cookies are assigned at random, one to each of the 24 treatment combinations. The 24 cookies are pressed (order randomized). Then one of the three oven temperatures is picked at random, the temperature of the oven set at that temperature, and all 8 cookies to be heated at that temperature are put in the oven. After 30 minutes of baking the 4 cookies assigned to the 30 minute baking time are removed. After another 10 minutes of baking the remaining 4 cookies are removed. This process is repeated for another 8 cookies out of the 24, with a different baking temperature. Then it is repeated for the remaining 8 cookies from the 24 at the last temperature. (the order of the three temperatures is randomized). Then a second batch of 24 ceramic cookies are made, and the above procedure (with independent randomization) is carried out. Finally, a third batch of 24 ceramic cookies are made and the procedure is carried out. At the end, there are a total of 72 hardness measures that are recorded, one for each of the 72 cookies. Set up the ANOVA table for the above experiment, showing sources of variation, degrees of freedom, and expected mean squares. Indicate which factors are fixed and which are random.

4) An economist was studying the relationship between the amount of money a family earns after taxes (Disposable Income), and how much they spend (Consumption). The economist took a random sample of 25 families, and for each family recorded disposable income and consumption. The economist fit a linear model relating these variables.

$$\text{Consumption} = a + (b \times \text{Disposable Income})$$

where a and b are constants. The constant b is called "the marginal propensity to consume." For the 25 families, the economist found that mean Consumption was \$ 18,000 ; and mean Disposable income was \$ 20,000. The standard deviation of consumption was \$1,000, while the standard deviation of Disposable Income was \$ 1,200. The (Pearson product moment) correlation between Consumption and Disposable Income was 0.80. (You may assume that the usual assumptions of the linear model hold, including homoschedasticity).

- a) Estimate the Average consumption for families whose disposable Income is \$15,000.
- b) Estimate the Standard deviation of consumption for families whose disposable income is \$ 15,000.
- c) Estimate the "marginal propensity to consume."

5) A supermarket buys 500 quarts of raspberries from a farm for 30 cents a quart. The supermarket must get rid of all the raspberries within one day, either by selling them to shoppers at 55 cents per quart, or by selling any leftover raspberries to a jam factory at 10 cents a quart. 20 percent of the days the shoppers buy exactly 400 quarts; 50 percent of the days they buy exactly 500 quarts, and the remaining 30 percent of the days they buy 600 quarts.

- a) What is the supermarket's average daily profit on raspberries?
- b) What is the standard deviation of the supermarket's daily profit on raspberries?

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Part I

Theory (90 minutes)

March 23, 1991

Instructions: Attempt any four out of the following five questions. Mention which four you would like to have graded. Open book and open notes.

1. Let X_i denote the number of meteors that collide with a test satellite during the i th orbit. Let $S_n = \sum_{i=1}^n X_i$; that is, S_n is the total number of meteors that collides with the satellite during n orbits. Assume that the X_i 's are independent and identically distributed Poisson random variables having mean λ .

(a) Find $E(S_n)$ and $\text{Var}(S_n)$.

(b) If $n = 100$ and $\lambda = 4$, find approximately $P[S_{100} > 440]$.

2. Let X and Y be continuous random variables having joint density f given by

$$f(x,y) = \begin{cases} \lambda^2 e^{-\lambda y}, & 0 \leq x \leq y \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find the marginal density of X and Y .

(b) Find the joint distribution function of X and Y .

3. Let $X_1, X_2, \dots, X_n$ be a random sample from the density

$$f(x, \theta) = \theta(1+x)^{-(1+\theta)}, \quad x > 0, \quad \theta > 0.$$

(a) Find the maximum likelihood estimator of $1/\theta$.

(b) Find the Cramer-Rao lower bound for unbiased estimators of $1/\theta$.

(c) Find the UMVUE of $1/\theta$ if such exists.

4. Let $X_1, X_2, \dots, X_n$ be a random sample of n from

$$f(X; \theta) = \theta^2 X e^{-\theta X}, \quad X > 0.$$

(a) Find a most powerful size α -test of $H_0: \theta = 1$ versus $H_1: \theta = 2$.

(b) Does there exist a uniformly most powerful size α -test of $H_0: \theta \leq 1$ versus $H_1: \theta > 1$? If so, what is it?

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Master's Comprehensive Part I (Continued)[3/23/91]

5. The heights of a large group of boys is normally distributed with mean μ_A and standard deviation 2.2 inches.

The heights of a large group of girls is normally distributed with mean μ_B and standard deviation 1.6 inches.

We seek to estimate $\mu_A - \mu_B$ correct within 0.1 inches with 95% confidence.

If we are going to take a simple random sample of n boys, and an independent simple random sample of n girls, how big must n be to achieve our goal?